Improving SAM-Road for Road Network Graph Extraction from Medium Resolution Sentinel Imagery in South Africa

Kelvin Wei
University of Cape Town
Cape Town, South Africa
wxxkel001@myuct.ac.za

Patrick Marais
University of Cape Town
Cape Town, South Africa
patrick@cs.uct.ac.za

Jonathan P. Shock
University of Cape Town
Cape Town, South Africa
jonathan.shock@uct.ac.za

Abstract

There is growing interest in utilising medium-resolution satellite imagery in monitoring road developments over time, as a cost-effective alternative to high-resolution images. However, road extraction from medium-resolution satellite imagery remains a challenging and under-explored field compared to high-resolution road extraction. The Sentinel satellite program freely provides multi-band medium-resolution satellite imagery. The generalisation from high-resolution to medium-resolution Sentinel imagery presents unique challenges attributed to lower resolution and multi-band domain shift. Recent advancements in models for high-resolution road extraction, such as SAM-Road, conceivably require modifications for Sentinel imagery. To address the additional bands and lower resolution of Sentinel imagery, we apply bicubic upsampling as a pre-processing step to the SAM-Road model and explore backbone and CNN-based model changes to its road segmentation component. Additionally, we introduce ROSA, a diverse dataset consisting of various biomes and urbanisation locations in South Africa. Evaluated on the ROSA dataset, our modifications to SAM-Road achieve 49.79 APLS, a 2.69 APLS improvement over our baseline SAM-Road model by utilising additional bands. Furthermore, to support practical road extraction across South Africa, we integrate our model into InstaGeo, an end-to-end geospatial machine learning framework.

Keywords

Road Graph, Satellite Imagery, Medium Resolution, South Africa

1 Introduction

Numerous studies highlight the need for more monitoring data to better manage the impact of roads. Roads can either positively or negatively affect the biodiversity of species, as they can act as both barriers and corridors for the movement of mammals and the spread of invasive plant species [20]. Furthermore, monitoring the development of road networks is crucial for understanding urban expansion, as road development leads to dynamic changes in land use and new infrastructure [18]. Because manual mapping of road networks is highly resource-intensive, deep learning-based extraction of road networks from satellite imagery is a promising approach to monitor road networks at scale [14, 25].

Most research is focused on accurate extraction of road networks from high-resolution imagery [9, 17]. While accurate road extraction is important, there are prohibitive costs associated with data access at a large scale. Cornebise et al. [4] report that a single very-high-resolution mosaic of Darfur was quoted at USD 4 million. In contrast, Sentinel-1 and Sentinel-2 programs freely provide medium-resolution (10m-30m) satellite imagery, which opens up possibilities in large-scale monitoring of historical road developments and changes. However, a primary challenge in utilising Sentinel imagery is attributed to its resolution, which hinders road extraction accuracy. The resolution entails that small-scale roads are commonly indistinguishable in 10 m imagery [13], limiting reliable road extraction to medium and large scale roads.

Recently, there have been promising advancements in high-resolution road extraction models. SAM-Road has shown state-of-the-art results in predicting road network graph topologies [9]. Additionally, its architecture supports high degrees of parallelism, making it suitable for large-scale road extraction from Sentinel imagery. This raises the question of how SAM-Road can be applied to Sentinel imagery.

The goal of this work is to modify the SAM-Road model for Sentinel-1 and Sentinel-2 imagery. The SAM-Road model consists of a road segmentation and graph prediction stage. Modifications target the road segmentation component to address challenges relating to decreased resolution and utilising multi-band inputs of satellite imagery. The road segmentation component utilises a foundational transformer model backbone, Segment Anything Model [15] (SAM), to segment roads from RGB imagery. However, SAM is designed for RGB imagery, which presents challenges when applied to multi-band satellite imagery. Additionally, transformer backbones may not be the most suitable for low-resolution segmentation due to a loss of local features during patching [3], whereas a CNN-based model preserves them. Therefore, we design our modifications to explore changing the SAM backbone to a domain-specific geospatial foundational backbone and replacing SAM-Road's road segmentation component with a CNN-based model.

Another constraint is a limited readily available dataset for road network extraction using Sentinel imagery. There are no publicly accessible road datasets for South African regions with Sentinel imagery. Commonly used high-resolution road datasets such as City-Scale [8] and SpaceNet [6] focus primarily on urban regions with limited diversity. As a result, we create a diverse SA road dataset across 9 biomes and differing urbanisation regions. Road labels are obtained from CDNGI, where roads are generated from high-resolution orthorectified aerial imagery and official ground surveys from the South African Government.

Models utilising satellite imagery usually have complicated data pipelines to process and clean imagery suitable for models, hindering practical deployment and utility. As a result, we also integrate our model into the InstaGeo framework to provide an interactive web application to extract end-to-end road networks from Sentinel imagery.

2 Background and Related Works

2.1 Medium-Resolution Road Extraction

The Sentinel-1 (S1) and Sentinel-2 (S2) satellites are part of the Copernicus programme led by the European Space Agency (ESA), and provide freely available medium-resolution multi-band imagery with a global revisit period of approximately five days since becoming fully operational in 2017 [12]. It is primarily challenging to reliably extract road networks due to insufficient spatial resolution. Fine-grained features like narrow roads are commonly visually indistinguishable [13] due to undersampling and mixed-pixel effects, where a single pixel captures several classes and reduces the spectral contrast. Therefore, most studies combine super-resolution and road segmentation techniques. Super-resolution complements road segmentation by ensuring that roads are not undersampled and can recover lost features [4, 22]. Super-resolution models trained with ground-truth high-resolution images are very promising, but bi-cubic upsampling is beneficial for road segmentation in the absence of high-resolution images. Ayala et al. [1] upsampled imagery by 4 times using bi-cubic interpolation before using a U-Net with a ResNet-34 [7] encoder for road segmentation. Mask labels were upsampled by re-rasterising road vector labels. The results show significant improvements in urban road segmentation, likely resulting from resolving pixel undersampling of sub-pixel roads.

The U-Net model [23] and similar variations are most commonly utilised as the road segmentation model with Sentinel satellite imagery. It utilises an encoder which captures high-level features and a decoder with skip connections to recover local features. CNN-based models, such as U-Net, struggle with road connectivity due to their inductive bias to prioritise local spatial features. Alternatively, transformers are suitable for capturing global contexts, but they may lose local features in the process of patching [3], and are not commonly utilised in isolation for road segmentation. Therefore, models like RNGDet++ [28] and SGCN [31] combine CNNs with transformers or graph convolutional networks (GCNs) to extract global context and preserve local features.

2.2 Road Graph Extraction

There is a recent trend in the shift from models predicting pixel representations to graph vector maps in road network extraction from high-resolution imagery. Road topologies are more naturally represented as graphs, and they provide better storage efficiency and accuracy [18]. Road graph extraction is broadly categorised into segmentation-based and graph-based approaches. Segmentation-based approaches [5, 14, 17] rely heavily on post-processing to improve the performance of an initial road segmentation mask. Even after post-processing, deriving a fully connected vectorised road network from segmentation masks is difficult [14]. Additionally, by utilising hand-crafted post-processing heuristics, they struggle with complex urban road structures such as multi-lane highways [9]. Graph-based approaches directly predict graph nodes and edges to form a road vector map. The complexities of urban regions are better represented due to an end-to-end approach, which eliminates hand-crafted heuristics with learnt features from data. While road graph extraction has largely been applied to high-resolution imagery, there is little work in extracting road vector maps from medium-resolution satellite imagery. In comparison to segmentation-based approaches, training graph-based approaches is difficult, and they are infrequently utilised, despite their advantages.

2.3 SAM-Road

SAM-Road [9] is a graph-based model with a split road segmentation and graph topology component. Road graphs are assumed to be $G = (V, E)$, where V is the set of road nodes (vertices), and E is the set of road lines (edges). SAM-Road primarily divides the task into vertex prediction and edge prediction, which are handled by the road segmentation and graph topology components, respectively. To predict road nodes, the road segmentation component generates road segmentation and intersection point masks, where NMS is applied to extract graph vertices. Road lines are predicted by the graph topology component, which utilises the extracted graph vertices and feature maps from the road segmentation component. The graph topology component consists of a bilinear sampler and a transformer-based graph neural network and is trained in a teacher-forcing manner. That is, ground-truth graph nodes are utilised rather than graph nodes from NMS of predicted road segmentation masks. The road segmentation component consists of an encoder-decoder model utilising a SAM encoder and a Fully Convolutional Network (FCN) decoder.

The model combines the advantages of segmentation-based and graph-based approaches to directly predict road graph networks. SAM-Road eliminates hand-crafted heuristics and supports high degrees of parallelism, making it suitable for large-scale road extraction. However, it also inherits disadvantages from segmentation-based approaches, where graph prediction heavily depends on the quality of segmentation results. SAM-Road uses a foundational transformer model backbone, the Segment Anything Model [15] (SAM). SAM is designed for RGB imagery, which presents challenges when applied to multi-band satellite imagery. There is also a domain shift in aerial or satellite imagery, which differs substantially from SAM's training data [27]. These challenges may prevent the road segmentation component from predicting quality segmentation masks and, as a result, lead to poor road graph predictions with Sentinel satellite imagery.

2.4 Existing Datasets

Large-scale Sentinel satellite imagery requires pre-processing and the management of additional bands and metadata, which is typically excluded in high-resolution RGB road datasets such as City-Scale [8] and SpaceNet [6]. Images of the Earth's surface are 2D projections of a 3D surface. A common coordinate system such as WGS84 can be utilised; however, they distort the imagery and road lengths depending on latitude and longitude coordinates. As a result, there exist many coordinate systems which minimise the distortions from projections based on the location. Sentinel images of South Africa span across multiple coordinate systems. Additionally, directly utilising individual satellite imagery from the CDSE STAC API [16] is challenging due to the presence of cloud cover and irregularly shaped images captured from a satellite's longitudinal orbital path. Issues arise from clouds and their shadows occluding road surfaces and from manipulating irregularly shaped images into rectangular images supported by deep learning models. Clouds are commonly handled with cloud masking or by creating

cloud-free mosaic imagery. Irregular-shaped images are handled by partitioning images into sub-patches.

There is a lack of Sentinel road extraction datasets, and Karnik et al. [13] provide the only publicly available Sentinel-2 road dataset. However, it covers only regions across India and is limited to RGB bands and pixel-based road masks. The WorldStrat [4] provides pairs of low-resolution (S2) and high-resolution images of SPOT 6/7 imagery. However, they primarily focus on super-resolution and do not provide road labels. Large-scale road graph extraction requires road graph labels and upsampling. Readily available datasets are not sufficient to support large-scale road graph extraction; hence, the motivation to create our own large-scale, diverse dataset across South Africa.

3 Methodology

This section outlines our methodology in extending SAM-Road for road graph extraction from Sentinel imagery situated in South Africa. We describe the construction of our dataset, then our pre-processing pipeline, SAM-Road modifications, and integration into InstaGeo. Prior work detailed in Section 2 informs our methods, and proposed modifications primarily build upon the original SAM-Road [9] paper and methods described in Ayala et al.'s [1] work in road extraction from Sentinel-2 imagery. Our methodology is presented to ensure reliability and reproducibility in the evaluation of our modifications.

3.1 Dataset Creation

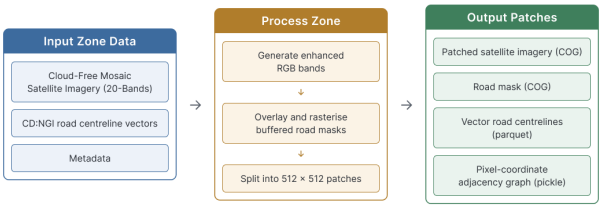


Figure 1: Workflow to generate image-label pairs for each zone-level image.

Our Roads of South Africa (ROSA) dataset was constructed using preprocessed zone-wide images with road labels from CDNGI and processed into a format suitable for model training. Figure 1 describes a simplified diagram of the processing steps involved per image to create the dataset. We conducted various pre-processing steps to enhance the quality of images, which were motivated by issues described in Section 2.4. We obtained cloud-free mosaics of satellite imagery from the Google Earth Engine (GEE) platform using the GlobalUrbanMapper Data Engine [32]. Mosaics were composited from multiple satellite images acquired between 1 January 2020 and 1 January 2021 to overlap with road label creation dates. The data engine provided preprocessed imagery of 20 bands composed of S2 Bands, DEM and related derivatives, urban masks and S1 SAR Bands. Bands were upsampled to 10m using bicubic upsampling in the case of a lower spatial resolution. Additionally, enhanced RGB bands were computed using CLAHE and gamma

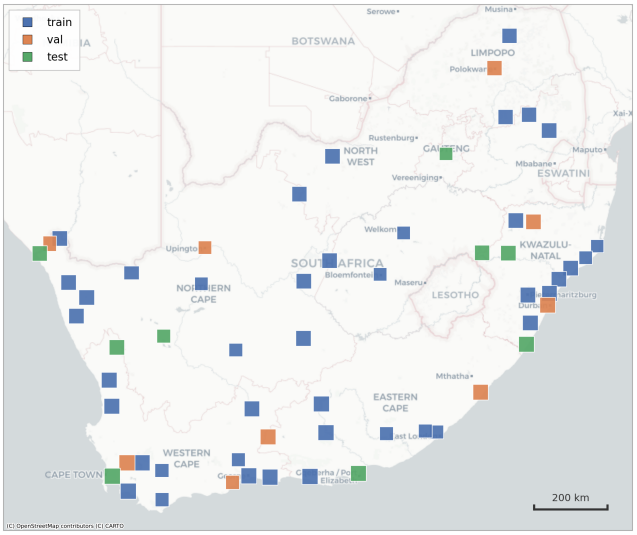


Figure 2: Sampled zone locations across South Africa.

correction to improve local contrast and enhance darker regions, as Karnik et al. [13] reported to improve road segmentation performance.

We generated rasterised road masks by aligning both the imagery and road centreline vectors. It was necessary to filter and remap road layers, as CDNGI provided a number of additional layers. Broadly, road labels were categorised according to their road width into large, medium and small. An approximate average road width for each category was chosen to rasterise buffered road centrelines at 10m resolution, shown in Table 7. We generated both road masks and graph labels to support re-rasterisation from vectors during upsampling and pixel-based or graph-based training.

Patches of 512 x 512 pixel satellite imagery were chosen to support random crops of 256 × 256 or 128 × 128 images. Corresponding road vectors and masks were generated by partitioning the image into a grid. Patches which contain no roads, empty pixel values and non-square patches were discarded during the patching process.

The ROSA dataset covers a diverse number of locations across biomes and urbanisation regions. A stratified sampling strategy was employed to ensure diversity, with manual selection and stratified random sampling. Urban regions contain the highest density of roads and were manually chosen across various biomes to ensure suitable locations. Peri-urban and rural regions were selected through stratified random sampling of a custom spatial strata mask containing approximately partitioned regions by biome, urbanisation and road presence. The custom mask was constructed using the VEGMAP [24] biome map, urban masks from ESA and WSF and CDNGI road labels. A sliding window operation with a 5000m window radius was utilised to obtain an urban density mask and thresholded at 0.2 and 0.5 to produce urban, peri-urban and rural regions. Then, we overlaid urbanisation regions, road labels and biome maps to further partition regions for stratified random sampling.

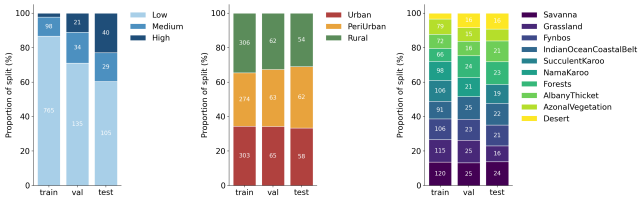


Figure 3: Composition of ROSA splits. Left to right: road density, urbanisation class, and biome.

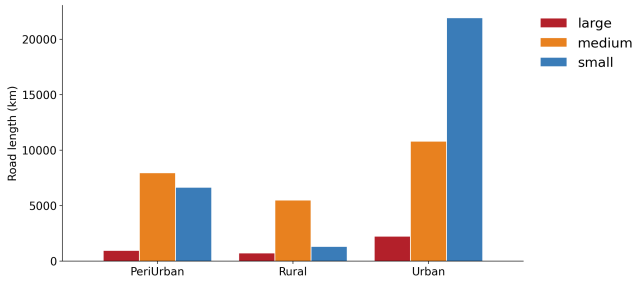


Figure 4: Bar graph showing the distances of roads across urbanisation.

It is noted that CDNGI contains labelling errors. Road labels in peri-urban and rural areas are focused on the main roads connecting built infrastructure, resulting in several unlabelled roads outside urban regions. Therefore, we have manually screened our test set to correct any errors. Primarily, we correct for missing and mislabelled roads. Sub-pixel roads were not discarded, as the model can potentially infer sub-pixel roads. We utilise QGIS to inspect our data on high-resolution Google Satellite imagery with Google Maps as basemaps for visual confirmation.

3.1.1 Characteristics. In total, our dataset contains 23 imagery bands with 1247 patches across 65 zones. Our dataset size is 11GB and the dataset folders and files are described in Table 6. Bands contain imagery from S1, S2, Urban masks from ESA, GISA, WSF and Enhanced RGB Bands. The sampled locations are shown in Figure 2. At a zone level, images were manually allocated to the validation and test sets to ensure a representative selection of diverse regions. Each zone is assigned either to the training (883 images), validation (190 images) or test set (174 images) to prevent data leakage. Metadata contains information at a patch level about road density, biome, urbanisation classification and mosaic date range. Distributions calculated from metadata are shown in Figure 3.

As shown in Figure 4 and 3, we have equally sampled urban and biome regions across our dataset. However, the dataset is largely characterised by small and medium scale roads, and there is an imbalance towards small-scale roads. The imbalance is largely due to the nature of urban regions and our sampling strategy. Small roads are concentrated in urban regions, such as dense residential streets. Meanwhile, large-scale roads are not effectively captured in our square zone samples. Square images do not capture a large proportion of lengthy roads exhibited by large roads. The road density graph in Figure 3 reinforces this finding, as most roads in peri-urban and rural regions are lengthy and sparse.

3.2 Pre-processing

Road vector labels were converted into a suitable format for training the SAM-Road model. Road labels were converted from a geospatial file format (geoparquet and gpkg) into a pickle file containing dictionaries of adjacency graphs and intersection masks. Additionally, the coordinate system of roads was converted from WGS84 to pixel-image coordinates.

On-the-fly upsampling and augmentations were implemented in a custom dataloader, heavily modified from SAM-Road's dataloader, to mitigate the effects of pixel undersampling and overfitting. We applied simple augmentations to enhance dataset diversity. Random crops of images are additionally applied with combinations of horizontal flips and 90-degree rotations. Road pixels occupy a small fraction of the dataset, which results in random crops containing empty roads. It was found that the imbalance of roads can lead to model collapse [2]. Hence, we discard and resample crops of images until upsampled road pixels exceed a small threshold of 50 pixels.

We followed Ayala et al. [1] as described in Section 2 and utilise bicubic upsampling, due to the lack of high-resolution images. We found that using bicubic upsampling of low-resolution road masks produced sampling artefacts of sub-pixel roads. Hence, we also re-rasterised road masks from road vectors and buffer values during upsampling. Additionally, intersection masks are computed from graph adjacency labels and similarly upsampled. Each training sample is a random 128 x 128 px crop of an image and upsampled with bicubic interpolation by 2x to 256 x 256 px with a spatial resolution of 5m. Although an upsampling factor of 4x would better sample roads with a 2.5m spatial resolution, it was not conducted due to resource constraints.

DEM bands and urban masks are available in our dataset, but not utilised as we focus on road extraction from Sentinel imagery. SAR descending bands from Sentinel-1 are also not utilised as they are not always available. We replaced the original RGB bands with enhanced RGB bands in training our models. As a result, we train on 12 bands, rather than all 23 bands generated.

3.3 Modifications to SAM-Road

Our method extended SAM-Road by modifying the road segmentation component of the model for Sentinel imagery. An overview of modifications is shown in Figure 5.

3.3.1 Backbone Replacement. TerraMind [11] is pretrained on multimodal Earth-observation data including S1 and S2 images. Our modifications explore substituting the SAM backbone with the TerraMind [11] backbone to determine the effectiveness of domain-specific pretrained models. Specifically, we replace the ViT encoder from the ViT-FCN encoder-decoder. We utilise the base variant, which has around 88M trainable parameters and is comparable to SAM's 80M trainable parameters.

3.3.2 Replacement with CNN-based model. We also explore the replacement of the segmentation component with a hierarchical CNN model to determine the effectiveness of local feature preserving methods. The models incorporate multi-scale feature pyramids

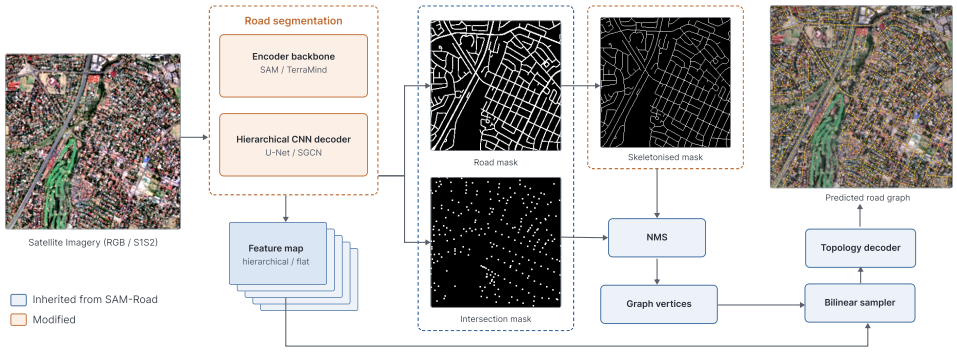


Figure 5: SAM-Road Modifications

and restore low-level features with skip connections. We utilise U-Net and SGCN. The U-Net model architecture is slightly modified from Ayala et al. [1] to use a ResNet-34 encoder initialised from ImageNet weights, and the first convolution is adapted for twelve input channels.

The SGCN model has a Graph Convolutional Network (GCN) as a bottleneck and has a global receptive field for global context, which combines the advantages of CNN and GCN. Additionally, because a hierarchical decoder exposes several feature maps rather than one, the vertex sampler bilinearly samples each map at the vertex location and concatenates the results.

3.3.3 Skeletonisation. A skeletonisation operation [30] was applied to road masks before NMS to assist in extracting graph vertices. We found that NMS picks inconsistent vertices from large buffer road masks generated from our dataset. This produces graphs whose edges zigzag across the roads. Therefore, to improve the quality of the road mask for NMS, we apply simple post-processing by filling holes smaller than 16 px, then reducing roads to a single-pixel width using the skeletonise operation. Skeletonisation preserves the topology of predicted roads, and the effect of skeletonisation is demonstrated in Figure 7. The graph decoder is also trained in a teacher-forcing manner. Hence, we found that skeletonisation produces NMS points more consistent with ground-truth road centrelines.

3.4 Integration with InstaGeo

Our model is integrated end-to-end with InstaGeo [29]. InstaGeo supplies acquisition and chipping of raw satellite imagery; however, they require cloud masks, which our models don't support. As a result, we additionally integrate GlobalUrbanMapper Data Engine [32] with the GEE Python package as an alternative source of data input to provide cloud-free mosaic imagery from which our models are trained. Predicted road graphs are served through InstaGeo's web map viewer, which renders the extracted centrelines over the source imagery in a browser.

3.5 Metrics

We primarily report graph-based metrics, but also include pixel-based metrics to evaluate both road graph extraction and road segmentation masks. Pixel-based metrics are used to evaluate SAM-Road's intermediate generated road masks. While graph-based metrics are correlated with the quality of road masks, it evaluates the connectivity of road graph extraction in addition to correctness.

3.5.1 Graph-based Metric. We use the Average Path Length Similarity (APLS) [6] to evaluate the topological correctness and connectivity of road graph extraction. Two random ground truth nodes and their corresponding prediction nodes have their optimal paths computed. The differences between their optimal paths are utilised to compute a score ranging from 0 (bad) to 1 (perfect).

3.5.2 Pixel-based Metric. Precision, recall and F1 are used to evaluate intermediate road segmentation masks. Precision focuses on the accuracy of predicted roads, while recall focuses on the model's ability to identify all ground truth roads. F1 is the harmonic mean of precision and recall, which balances the precision and recall scores.

$$\mathrm{P} = \frac{TP}{TP + FP}, \qquad \mathrm{R} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2\,\mathrm{P}\,\mathrm{R}}{\mathrm{P} + \mathrm{R}}.$$

3.6 Experimental Framework

As our baseline, we slightly modify the original SAM-Road model with skeletonisation as described in Section 3.3.3 to mitigate the effect of different road-scale buffers. The loss function is also modified from vanilla binary cross-entropy loss to an equally weighted BCE and Dice loss [18, 19] commonly used to handle road class imbalance. All models utilise upsampling and augmentations described in Section 3.2, with skeletonisation and the modified loss function for consistency.

We run two experiments to isolate the effects of additional bands against our road segmentation component modifications. Firstly, we train our models with enhanced RGB bands to compare the original SAM-Road baseline against our backbone and CNN-based model replacements. In the second experiment, each model is trained with 12 bands consisting of enhanced RGB bands combined with S1 and S2 (S1S2) bands. The original SAM-Road model is excluded from

Table 1: Model benchmarking results on the ROSA test set for RGB and S1S2 input. Models with ± indicate the mean and standard deviation over 3 training seeds. Otherwise, results are over a single seed. The best results are bolded.

Input	Model Variant	Pretrain	Param (M)	FLOPs (G)	Precision	Recall	F1	APLS
RGB	SAM-Road	SA-1B	87.21	69.37	61.31 ± 0.71	**60.06 ± 0.24**	**58.78 ± 0.40**	**47.10 ± 0.43**
	U-Net	ImageNet	**24.90**	**15.65**	59.71	56.70	56.05	45.43
	SGCN	–	44.36	151.74	**63.90**	54.88	57.33	45.41
	TerraMind	TerraMesh	86.41	44.27	57.34	58.15	55.73	42.87
S1S2	SGCN	–	44.37	152.42	**65.56 ± 0.48**	60.17 ± 0.72	**61.04 ± 0.37**	**49.79 ± 1.55**
	U-Net	ImageNet	**24.93**	**16.57**	62.94 ± 0.76	**60.26 ± 0.67**	59.85 ± 0.22	48.92 ± 0.17
	TerraMind	TerraMesh	88.18	88.66	62.09	59.62	58.65	45.88

this experiment, as it does not support S1S2 bands. Rather, the effect of additional bands can be compared against the corresponding RGB models from the first experiment.

We retrain the original SAM-Road model and the 2 best models with 3 random seeds. The mean and standard deviation are reported to account for the random variability in training. Statistical comparisons utilise a non-parametric two-sided paired Wilcoxon signed-rank test [26], which statistically tests the difference between two models in performance on the test set. A non-parametric test is used as we don't assume our data follows a specific distribution. In performing the statistical test, we paired each model's evaluation metrics per image, which are respectively averaged across 3 seeds.

Our initial tuning experiments yielded minimal or negative improvements over the hyperparameters reported for SAM-Road. Due to time and resource constraints, we fixed the training hyperparameters utilised across all model modifications, which is an area for improvement. An Adam optimiser with a base LR of 0.001 and a pre-trained image encoder LR of 0.1× the base LR was chosen from SAM-Road, and we trained our models at a batch size of 16 for 100 epochs.

During inference, we follow the SAM-Road paper [9] to tune thresholds for binary classification tasks on the validation set and perform sliding window inference in extracting roads from the test set.

3.7 Ethics

This project requires no formal ethics clearance as it relies on publicly available medium-resolution satellite imagery and road annotations. Medium-resolution images have insufficient resolution to identify individuals, and we limit our area of study to South Africa. Utilised models and tools were open-source or under an academic licence.

4 Experiments and Results

This section discusses the main results of our experiments relating to our modifications to SAM-Road. We discuss our findings with RGB and S1S2 bands, the performance efficiency of our models and ablation results.

4.1 RGB SAM-Road Baseline

The baseline SAM-Road model yielded an average APLS score of 47.1% and 58.78% F1 score from its intermediary road masks over 3 seeds (Table 1). Despite the dissimilarity between RGB satellite imagery and natural photographic images used in SAM's pretraining [15], the model effectively handles the domain shift to RGB Sentinel-2 imagery with full fine-tuning and bicubic upsampling. We find that it performs particularly impressively in the recall of roads in its road segmentation masks. The baseline SAM-Road model utilises a SAM backbone, and its pretrained weights appear to be beneficial at detecting general road-like features and assisting in recall. This suggests that SAM's pretrained weights have impressive generalisation capabilities in detecting road-like features from pretraining.

In the RGB setting, the baseline SAM-Road is the best model overall, and our modifications to SAM-Road underperform the baseline. This is a surprising result given our assumption, from existing literature, that ViT lose fine-grained features. However, it showcases the impressive generalisation capabilities of pretrained models. While we did modify the original SAM-Road model as described in Section 3.6 for a fair comparison, the result shows that our modifications to utilise domain-specific foundational models and CNN-based models do not improve performance.

4.2 Effect of Additional Bands

Across every model, S1S2 input outperforms its RGB counterpart, indicating that the additional bands contribute positively to road extraction. The gains are consistent but modest relative to the RGB baseline. Results suggest that enhanced RGB bands are the major driver of performance, and additional S1S2 bands strengthen predictions. Performance with S1S2 is likely limited by spatial resolution. RGB, NIR, and SAR bands are 10m, whereas additional S2 bands are 20m and upsampled to a common resolution.

We find that generally improvements occur in vegetation-dense and urban areas with high buildings. While we can not definitively interpret the improvements, we hypothesise that S1S2 bands carry information about physical surface changes and interactions with vegetation and water not present in RGB bands. Our findings are consistent with existing literature, as empirical results have found S1S2 Bands to be effective in Land-Use and Land Cover (LULC) for mapping built-up areas, water and forests [10].

Figure 6 contrasts each model's results. The models with S1S2 input usually perform better than RGB counterparts. Interestingly, we find that S1S2 input can sometimes also hinder road detection in farm-like regions such as the Coastal Belt. Common errors occur

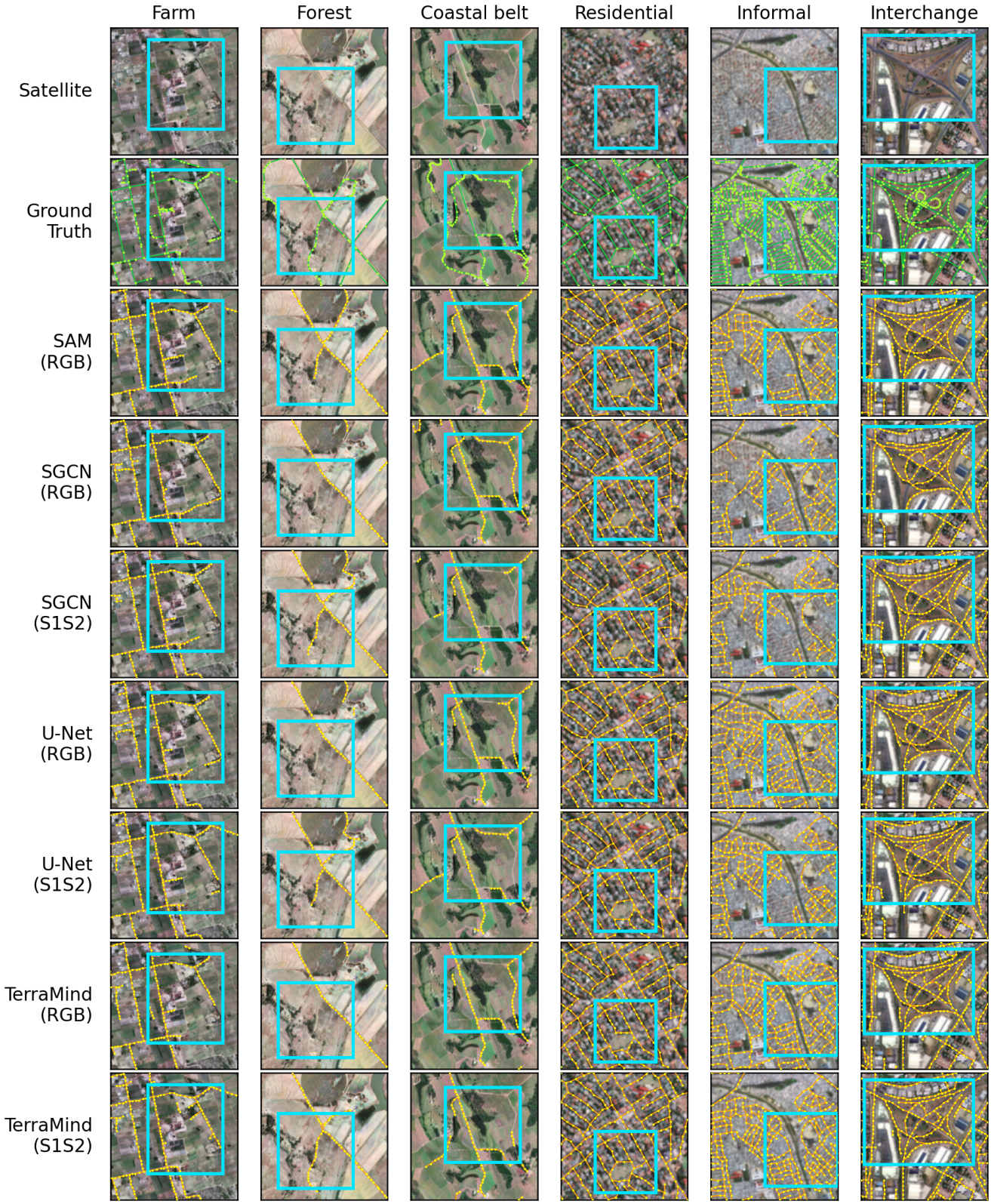


Figure 6: Side-by-Side Model Comparisons

Table 2: SGCN and U-Net results stratified by urbanisation categories. Metrics are averaged over 3 training seeds, with mean ± standard deviation. p-values are computed using paired Wilcoxon signed-rank tests. Results are bolded where the difference is significant at $\alpha = 0.05$.

			Precision		Recall		F1		APLS	
Urbanisation	Patches	Model	Value	p	Value	p	Value	p	Value	p
Urban	58	SGCN	**70.37 ± 0.49**	8.9×10^{-10}	69.00 ± 0.91	0.88	**69.39 ± 0.33**	7.2×10^{-9}	**63.62 ± 0.61**	2.5×10^{-4}
		U-Net	65.62 ± 0.64		69.24 ± 0.37		67.08 ± 0.49		62.28 ± 0.46	
PeriUrban	62	SGCN	**65.32 ± 0.83**	1.5×10^{-5}	56.86 ± 0.68	0.34	**59.49 ± 0.50**	4.3×10^{-3}	42.65 ± 1.30	0.069
		U-Net	62.62 ± 0.88		57.08 ± 1.50		58.52 ± 0.45		41.21 ± 0.81	
Rural	54	SGCN	60.67 ± 2.25	0.56	54.48 ± 0.82	0.40	53.86 ± 0.78	0.54	42.95 ± 3.16	0.33
		U-Net	60.43 ± 1.53		54.28 ± 0.57		53.63 ± 0.60		43.38 ± 0.47	

in highway interchanges, small-scale roads and the global connectivity of roads. Models also appear to perform poorly and overfit the informal regions. CDNGI labels commonly mistakenly exclude small blocks of roads in urban regions, such as informal settlements. Models can be susceptible to overfitting on undesirable characteristics of the ROSA dataset. However, TerraMind and baseline SAM predictions appear to have more coherent road structures and are less susceptible to overfitting.

4.3 Segmentation Component Analysis

Our results show that our road segmentation component modifications underperform against the baseline RGB SAM-Road model. However, we find that our modifications can utilise additional S1S2 bands and display noticeable improvements in performance. SGCN with S1S2 bands achieves the highest APLS score, but has a relatively high variability across different training seeds. SGCN's performance is attributed to high precision. Despite the highest APLS score, the worst-performing SGCN seed achieved 48.38 APLS, and the corresponding U-Net model trained on the same seed achieved 48.77 APLS. As a result, we can not definitely conclude whether SGCN or U-Net is our best model. However, we chose the SGCN model as our best result, as the best-performing seed achieves 51.45 APLS.

4.3.1 SGCN. Strangely, we find SGCN performs poorly when restricted to RGB input. We suspect the reason to be a lack of pretraining. SGCN is the only model without pretrained weights and is more susceptible to overfitting our dataset due to training from scratch. We find the models with pretrained weights typically have a higher recall than precision. SGCN has one of the lowest recall scores, but its performance is attributed to its higher precision. This is consistent with the model's lack of pretraining, where the model doesn't generalise well to unfamiliar roads and overfits on familiar roads. The trade-off between precision and recall is beneficial with S1S2 bands, as the additional information causes a substantial increase in precision without largely affecting recall negatively, but leads to poor results with RGB bands. The high standard deviation entails that the model is more influenced by random augmentations and initialisation during training.

SGCN shows an inconsistency in its low trainable parameters but high FLOPs during inference, making it the most computationally expensive model. We were unable to reproduce the original paper's [31] reported 60 GFLOPs. The high computational cost is largely due to the model's modifications to its ResNet-50 backbone, which maintains high-resolution feature maps with dilated convolutional layers.

4.3.2 U-Net. The U-Net is the most performance-efficient model. The model achieves the second-best performance despite having the lowest trainable parameters and computational complexity.

4.3.3 TerraMind. Surprisingly, a domain-specific geospatial foundational model performs poorly. While we find that the domain-specific pretraining assists with recall, TerraMind underperforms in precision. TerraMind's pretraining is largely focused on correlation learning and multimodal generation. Additionally, TerraMind's architecture is more susceptible to the loss of fine-grained features due to tokenising and patching of images. Therefore, pretraining is helpful, but TerraMind loses fine-grained features due to a mismatch between its pretraining objective and its architecture in preserving fine-grained features.

4.3.4 Urbanisation and Zone-Level Breakdown. Table 2 shows a breakdown of U-Net and SGCN results with S1S2 bands. The breakdown of metrics is across urban, peri-urban and rural regions. We also conduct statistical significance testing on our results between the two models and report the p-value.

In comparison to rural and peri-urban regions, our models perform the best in urban regions by a large margin. The relatively high performance in urban regions is attributed to the standardised infrastructure and denser interconnected roads, which lead to fewer APLS penalties due to road disconnections. The challenge with urban regions is primarily due to the dense and complex road structures. Rural regions primarily face sub-pixel roads, while peri-urban regions have complex road structures and sub-pixel roads. As noted in Section 3.1, our dataset prioritises roads connecting built infrastructure, which is a challenge for our models as they independently predict image sub-patches, reducing global context.

Our results show that SGCN is statistically better on precision in urban and peri-urban contexts. However, while we find SGCN to perform better on APLS in urban regions, it is not also significantly

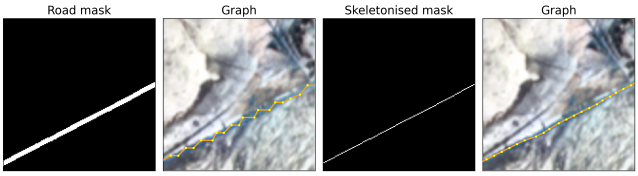


Figure 7: Skeletonisation Comparison

better in peri-urban regions. Table 5 provides a finer breakdown across the nine zones in our test set. It further shows that the models perform the worst in regions with sparse vegetation, like the Karoo and desert.

4.4 Speed and Performance

CNN-based models are able to utilise additional bands without a substantial increase in computational complexity and trainable parameters. Additional bands are handled by adjusting the input channels from the first convolutional layer. In comparison, transformers such as TerraMind require a substantial increase in parameters and computational complexity to handle additional bands. The trainable parameters and FLOPs per 256x256 input image of our models are shown in Table 1.

SAM-Road utilises a sliding window for inference. A sliding window is a common technique to improve performance. A sliding window can trade speed and performance during inference by adjusting the overlap of sliding windows. This trade-off can be seen in Table 4 over single-seed evaluations of SGCN, U-Net and Baseline SAM-Road models.

A sliding window with overlap is found to be essential for the connectivity of graphs. In the 4x4 setup, there is no overlap in the windows, and we see that the APLS drops drastically. The model predictions are worse near the edge, and graph disconnections occur frequently as a result.

Inference was conducted with a single GTX1060 GPU. Even with an old GPU, the models generate predictions for a single 512x512 image in less than 30s.

4.5 Ablation

We conduct an ablation of our model to determine the effect of skeletonisation on RGB and S1S2 bands with our best model variation (SGCN) over a single seed. Additionally, SAM-Road differs from segmentation-based models due to its graph decoder head, which replaces hand-crafted post-processing heuristics. Therefore, we also conduct an ablation with SAM-Road's intermediary road masks to compare the performance of segmentation-based post-processing against SAM-Road's graph decoder head. We utilise the post-processing scripts from the CRESIv2 paper [5]. Our results are described in Table 3. We compute ablation results stratified over urbanisation regions and the whole test set and show statistical significance results. Paired statistical tests are conducted to determine the statistical significance of our skeletonisation modification over segmentation-based model post-processing and skeletonisation not applied to road masks.

We find that applying skeletonisation to the predicted road mask before NMS graph vertex extraction improves metrics. Figure 7 showcases the effect of skeletonisation. Thick road segmentation masks result in a zigzag pattern. Performance gains from skeletonisation modification result from better rural and peri-urban region scores. Furthermore, we find that graph-based approaches perform better in urban regions, as they significantly outperform post-processing results. However, post-processing significantly improves rural and peri-urban results. Our findings are consistent with the original SAM-Road's results over urban regions as discussed in Section 2.3, but post-processing is required to improve performance outside urban regions.

4.6 Discussion

Our results support the feasibility of road graph extraction from medium-resolution Sentinel imagery. The strongest factor in performance improvement is utilising additional Sentinel-1 and Sentinel-2 bands over enhanced RGB bands. In contrast, our road segmentation component modifications do not improve upon the baseline SAM-Road. The contribution of our modification is twofold. First, our skeletonisation modification improves performance, and secondly, U-Net models with S1S2 input segment roads better than SAM and at negligible cost. The SGCN model with S1S2 input is our best model, and its performance is attributed to improvements in precision, but it has a high computational cost with relatively high variability.

Expectedly, our metrics fall short of high-resolution road extraction scores of 60-70% APLS [18] and 70-80% [25] F1 found in the literature. While these metrics are not directly comparable, they demonstrate that road extraction accuracy is sacrificed in utilising freely available medium-resolution Sentinel imagery. Our dataset contains a large proportion of small-scale roads which require high-resolution imagery for reliable road extraction. Small-scale road extraction is possible from Sentinel imagery, but we find the results to be unreliable with our current methods.

It is a common issue that road extraction models struggle to maintain the connectivity of roads, where graph-based methods are more suitable. However, SAM-Road inherits the disadvantage of pixel-based segmentation models, as it relies on road segmentation masks, which are known to have issues in maintaining road connectivity. Similarly, we find our models struggle to maintain global connectivity. However, models are also influenced by undesirable road labelling characteristics of our training dataset. Detecting major roads connecting built-up infrastructure is difficult as models predict independently on small patches.

From our ablation, we find that SAM-Road's graph decoder improves performance in urban regions. However, it struggles outside urban regions and is outperformed by segmentation-based post-processing. End-to-end graph extraction has the advantage of learning features rather than hand-crafted heuristics, but learnt features become suboptimal with simpler road structures in rural and peri-urban regions. Therefore, we find that our skeletonisation post-processing aids the model by constraining the predicted graph vertices towards the centre of roads.

5 Limitations

We acknowledge that the labels of our datasets are not perfect, and only the test set is manually inspected and cleaned. Additionally,

Table 3: Effect of skeletonisation and advanced graph post-processing on SGCN. S1S2 rows are averaged over 3 training seeds (mean ± standard deviation). Otherwise, results are over a single seed. p-values are computed using paired Wilcoxon signed-rank tests. Values are bolded when they are better and differ significantly.

Variation	Graph construction	Whole Test Set	Urban	Peri-Urban	Rural
S1S2 Bands	Unmodified	48.89 ± 1.24	63.62 ± 0.23	41.47 ± 1.31	41.37 ± 2.81
	NMS + Skeletonisation	49.79 ± 1.55	$\mathbf{63.62 \pm 0.61}$	42.65 ± 1.30	42.95 ± 3.16
	CRESI post-processing	50.56 ± 0.80	62.61 ± 0.97	43.67 ± 0.90	$\mathbf{45.47 \pm 0.93}$
	p (unmodified vs. skeletonisation)	1.1×10^{-6}	0.46	1.7×10^{-4}	9.0×10^{-6}
	p (skeletonisation vs. CRESI)	0.058	4.0×10^{-5}	0.19	2.6×10^{-7}
RGB	Unmodified	44.90	55.86	39.61	39.09
	NMS + Skeletonisation	45.41	**55.57**	40.94	39.53
	CRESI post-processing	46.33	52.63	**43.78**	**42.35**
	p (unmodified vs. skeletonisation)	7.9×10^{-3}	0.49	4.3×10^{-3}	2.6×10^{-2}
	p (skeletonisation vs. CRESI)	0.51	5.6×10^{-6}	5.3×10^{-3}	6.9×10^{-4}

Table 4: Table describing the time cost and accuracy over varying levels of sliding window strides

Model	Setup	Time Cost (s/image)	APLS
SGCN	16×16	27.47	51.45
	8×8	7.20	49.60
	4×4	2.13	23.53
U-Net	16×16	5.07	49.11
	8×8	1.61	48.09
	4×4	0.77	19.65
SAM-Road	16×16	17.03	47.47
	8×8	6.35	45.38
	4×4	2.21	21.24

our dataset largely consists of small and medium scale roads, and urban images contain the largest proportion of roads by distance. We have not accounted for the skewness of the dataset, and it may hinder a model's generalisation capabilities. Additionally, our work upsamples images by 2x to 5m. Most studies utilise 4x upsampling to 2.5m and would better resolve pixel undersampling of roads.

The creation of our dataset utilised single yearly mosaic images, which discards quarterly seasonal variations, particularly noticeable in vegetation-dense regions. Furthermore, our assumption that roads infrequently change over a year may not hold over rural and peri-urban regions. As a result, yearly mosaic images may distort frequently changing roads.

Our models are also limited in reliably detecting small-scale roads and maintaining connectivity, as global context is limited due to SAM-Road's architecture to independently predict small sub-image patches.

6 Conclusion

We investigated modifying SAM-Road, a high-resolution road graph extraction model, to medium-resolution Sentinel imagery. To support this, we introduced ROSA, a diverse road extraction dataset spanning across nine South African biomes and three urbanisation categories, built from CDNGI road labels and cloud-free Sentinel mosaics. The dataset artefacts were created to support re-rasterisation during upsampling and graph-based model training.

We evaluated SAM-Road modifications involving backbone replacement, CNN-based model replacement, and a skeletonisation post-processing operation. For fair comparisons, we apply bicubic upsampling as a pre-processing step and skeletonisation as an internal post-processing step. These modifications are found to improve performance in previous studies or in our results. Our results with RGB input demonstrate that our backbone replacement and CNN-based model replacements do not improve the SAM-Road model. However, we find that additional Sentinel-1 and Sentinel-2 bands do improve performance, as our modification with additional bands outperforms the baseline SAM-Road model with SGCN and U-Net models.

Our ablation results suggest the SAM-Road model is suitable in urban regions. However, segmentation-based methods with post-processing would outperform SAM-Road more rural regions.

Additionally, we integrated our models into InstaGeo to provide an end-to-end pipeline from data acquisition to road network extraction and visualisation from the browser. Our performance results with SAM-Road and our variants demonstrate that the models can run on modest hardware.

6.1 Future Work

Future work should include improving the dataset and exploring super-resolution with high-resolution ground truth imagery, where time-series images can be utilised to take advantage of the high revisit rate of the Sentinel satellite. Studies have found that super-resolution with multiple images is more effective than single-image super-resolution [4]. This requires expanding the dataset with multiple low-resolution pairs with a single super-resolution image.

Additionally, SAM can be adapted to utilise additional S1 and S2 bands [21] to improve upon its existing segmentation capabilities. Furthermore, combining fine-grained features and global features is a promising direction to both accurately extract road networks and preserve road connectivity.

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7 AI Statement

Generative AI models (Claude, Gemini) were used to review drafts of this paper for grammatical errors, clarity and constructive feedback. All feedback was critically evaluated before being incorporated into the writing.

A Supplementary Information

Table 5: SGCN and U-Net stratified by zone, averaged over 3 training seeds (mean ± standard deviation).

Biome	Location	Category	Patches	Model	Precision	Recall	Road F1	APLS
Fynbos	Cape Town	Urban	21	SGCN	**70.62 ± 0.84**	**69.50 ± 1.94**	**69.80 ± 0.74**	**67.02 ± 1.26**
				U-Net	64.88 ± 0.71	69.49 ± 1.04	67.01 ± 0.88	66.26 ± 0.89
Grassland	Johannesburg	Urban	16	SGCN	**66.15 ± 0.58**	69.96 ± 0.66	**67.83 ± 0.12**	**63.33 ± 0.97**
				U-Net	59.85 ± 1.11	**70.25 ± 0.24**	64.44 ± 0.60	59.88 ± 0.56
Albany Thicket	Port Elizabeth	Urban	21	SGCN	**73.33 ± 0.31**	67.78 ± 0.34	**70.17 ± 0.10**	**60.43 ± 0.76**
				U-Net	70.77 ± 0.25	**68.22 ± 0.35**	69.16 ± 0.18	60.13 ± 0.28
Savanna	Random Sample	Peri-Urban	24	SGCN	**66.94 ± 0.46**	**65.15 ± 0.28**	**65.54 ± 0.18**	**47.30 ± 0.34**
				U-Net	63.37 ± 1.29	63.70 ± 1.14	62.94 ± 0.23	45.02 ± 0.35
Forests	Random Sample	Rural	23	SGCN	**71.85 ± 2.88**	**53.65 ± 1.38**	**59.09 ± 1.04**	**46.17 ± 2.54**
				U-Net	70.87 ± 1.25	52.78 ± 0.75	58.17 ± 0.65	43.08 ± 0.95
Indian Ocean Coastal Belt	Random Sample	Peri-Urban	22	SGCN	**69.08 ± 0.79**	52.85 ± 0.49	**58.95 ± 0.14**	**43.66 ± 1.09**
				U-Net	66.20 ± 0.71	**54.20 ± 0.85**	58.74 ± 0.46	41.17 ± 0.72
Succulent Karoo	Random Sample	Rural	19	SGCN	53.78 ± 0.37	**64.25 ± 0.62**	55.19 ± 0.46	42.76 ± 1.06
				U-Net	**55.86 ± 2.40**	63.69 ± 0.93	**55.77 ± 1.06**	**47.13 ± 2.01**
Azonal Vegetation	Random Sample	Rural	12	SGCN	**50.16 ± 5.20**	40.61 ± 0.61	**41.73 ± 0.82**	37.37 ± 8.82
				U-Net	47.66 ± 4.09	**42.28 ± 0.69**	41.54 ± 1.02	**38.02 ± 3.15**
Desert	Alexander Bay	Peri-Urban	16	SGCN	**57.72 ± 3.04**	49.92 ± 1.73	51.16 ± 2.03	34.31 ± 4.43
				U-Net	56.59 ± 1.43	**51.08 ± 3.79**	**51.58 ± 1.87**	**35.54 ± 2.11**

Table 6: Contents of the ROSA dataset.

Path	Format	Notes	Size
`{split}/imagery/`	GeoTIFF	23 bands, float32, 512 × 512 @ 10 m	10.3 GB
`{split}/masks_raster/`	GeoTIFF	binary road mask, uint8	157 MB
`{split}/mask_keypoints/`	GeoTIFF	binary intersection mask, uint8	157 MB
`{split}/masks_graph/`	GeoParquet	road vector centrelines in WGS84 coordinates	168 MB
`{split}/mask_adj_graphs/`	pickle	graph adjacency lists in image pixel coordinates	168 MB
`splits/*.csv`	CSV	truncated metadata for data loader	768 KB
`metadata.parquet`	Parquet	detailed metadata per image	139 KB
`norm_stats.yaml`	YAML	precomputed mean and std over train set	714 B
`dataset_summary.yaml`	YAML	stratification proportions over split sets	1.8 KB
Total			11 GB

Table 7: Table of CDNGI Road Classification and Rasterisation Buffers

Scale	CDNGI class	Buffer (m)
Large	National Freeway	8.0
	National Road	8.0
	Arterial Road	8.0
	Main Road	8.0
Medium	On/Off Ramp	6.0
	Secondary Road	5.0
	Other Road	5.0
Small	Street	4.0
	Track	3.0
	Slipway	3.0
Ignored	Footpath	—